

MATTHEW DAVIES

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Software Developer | Backend Systems, Cloud & AI

EDUCATION

University of Windsor

Honours Computer Science | AI Specialization | Math Minor

Windsor, ON

Sep. 2022 – Aug. 2026

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, Golang, C++, C, C#, PHP

Web / Full-Stack: FastAPI, Spring Boot, Node.js, React, REST APIs, CSS, MySQL, PostgreSQL, Laravel

Automation / Tools: Selenium (Python), Git, GitHub, Linux, Bash/Shell Scripting, Postman, JUnit, xUnit

Cloud / DevOps: AWS (EC2, RDS, Elastic Beanstalk), Docker, CI/CD, Kubernetes (fundamentals)

Data / Analytics & ML: Retrieval-Augmented Generation (RAG), FAISS, embeddings, SQL, ETL pipelines, PyTorch, Pandas, NumPy, scikit-learn, Matplotlib

Systems / Performance: CUDA, GPU Programming, Parallel Computing

PROJECTS

CS Tutor Bot | RAG-Based Course Assistant

- Built a **Retrieval-Augmented Generation (RAG)** agentic AI system for grounded document understanding and question answering.
- Implemented data ingestion, embedding generation, and semantic retrieval using **MiniLM + FAISS**.
- Deployed a **FastAPI** backend supporting **125+ students** with source-aware, non-hallucinating responses.
- Designed and validated end-to-end **data pipelines and backend services** for ingestion, indexing, retrieval, and response generation with a focus on correctness and reproducibility.
- Architected a cloud-ready AI service with modular ingestion, retrieval, and inference components designed for scalable deployment.

Spring Boot Microservice | Dockerized & Deployed to AWS

- Designed and built a production-ready **Spring Boot REST API** using **Java 17, JPA, and PostgreSQL**.
- Implemented JPA entities, repositories, and CRUD endpoints for task and user management.
- Containerized the service using **Docker** with a custom Dockerfile and multi-stage build for optimized images.
- Deployed the application to **AWS Elastic Beanstalk**, configuring Procfile routing, and EC2 platform settings.
- Debugged cloud deployments using **AWS logs, EC2 instance, network configuration, and REST clients**.
- Tested REST API endpoints using Postman and debugged deployment issues through AWS logs and application diagnostics.

Full-Stack Quote Generator | React + FastAPI + PostgreSQL

- Developed a full-stack web application using **React (TypeScript)** and **FastAPI** to create, store, and display user-submitted quotes.
- Built RESTful backend services with **SQLAlchemy + PostgreSQL**, implementing validation and API design.
- Handled schema design, API routing, UI development, and database integration end-to-end.

Google Docs API | Data Extraction & ETL Pipeline

- Developed a **Python ETL pipeline** using the **Google Docs API** to extract, transform, and load structured data from a private document.
- Parsed and validated semi-structured table data, handling inconsistent formats and ensuring data integrity.
- Transformed coordinate-based symbol data into a 2D grid to reconstruct and visualize a decoded message.

Hypergraph Neural Networks for Node Classification | Representation Learning Research

- Designed and implemented multiple **Hypergraph Neural Network (HGNN)** models for node classification, extending traditional GNNs to capture **higher-order relationships** via hyperedges.
- Developed **dynamic hypergraph learning methods** that iteratively update hyperedges based on learned node embeddings, enabling adaptive structure learning during training.

EXPERIENCE

NASA Space Apps Challenge

Software Developer | Hackathon

Windsor, ON

Oct. 2025

- Worked in a multidisciplinary team to build an interactive web platform for exploring NASA's high-resolution imagery.
- Implemented UI functionality and image manipulation tools using **React (TypeScript)** and **Go**.
- Collaborated with cross-functional teammates to translate user needs into data-driven UI and backend functionality.